Sora Kim

Practitioner-Researcher in Regulatory HCI

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Research

Research Program

My research investigates how AI can be made useful in high-stakes regulated environments — pharmaceutical manufacturing, aviation, clinical care, finance — without compromising the regulatory structures that make those environments trustworthy. Working as a design researcher embedded in regulated work, I develop two interlocking lines of inquiry: AI systems engineered to operate within regulatory constraints, and the educational scaffolding required for practitioners to deploy them responsibly.

Research Interests

Regulatory HCI · AI in regulated work · Judgment-preserving AI design · AI adoption in high-stakes environments · Practitioner-AI interaction · Design for regulated environments · Research through Design · AI literacy and educational scaffolding

Industry Research Experience

GMP-Regulated Pharmaceutical Manufacturing Site Seoul, South Korea
Quality Control / Quality Systems September 2022 – Present

- Embedded practitioner-researcher in regulated biomanufacturing quality operations (∼4 years); sustained ethnographic observation of procedural document use, deviation handling, corrective action workflows, and regulatory audit preparation.
- Designed and deployed **twelve productivity tools** serving Quality Control operations (see Deployed Tools); observational data and fieldwork form the empirical foundation of the research program.
- Designed and delivered a differentiated AI training program for the Quality Control team (managers: tool-building; operators: AI-assisted authoring), motivating the educational artifacts below.

Publications

Seven working manuscripts across two tracks. PDFs available on request.

Organizational Theory & Science and Technology Studies

A four-paper lifecycle theory: pre-action structuring (AAC), procedural document infrastructure (CF), investigative convergence (RC), AI-accelerated closure (PAC).

Kim, S. (2026). *Anticipated Accountability Convergence: How Organizations Structure Responsibility Before Action.* Working paper.

Kim, S. (2026). *Cognitive Friction: Procedural Documents as Accountability Infrastructure in Regulated Organizations.* Working paper.
Companion code: `sorakim-lab/sop-cognitive-friction-pilot`.

Kim, S. (2026). *Responsibility Convergence: How Investigative Systems Stabilize Attribution.* Working paper.

Kim, S. (2026). *Premature Accountability Convergence: How Technology Accelerates Responsibility Stabilization in Organizations.* Working paper.
Companion code: `sorakim-lab/fda-warning-letters-pac-analysis`.

Computational Social Science

Three computational papers operationalizing accountability across regulatory domains; cross-domain replication validates the lifecycle theory developed on FDA data.

Kim, S. (2026). *Responsibility as Multi-Layer Signal: A Computational Framework for FDA Warning Letters.* Working paper.
Synthesizes four computational components, each with a companion repository: `premature-convergence-detection`; `accountability-graph-extraction`; `responsibility-attribution-prediction`; `responsibility-distribution-metric`.

Kim, S. (2026). *Designed Attribution: Coding-Level and Grammatical-Level Evidence of Organizational Accountability Displacement in NTSB Aviation Investigations.* Working paper.
Replication code: `sorakim-lab/ntsb-attribution-analysis`. Two-phase analysis: structured coding (n=71,114) and dependency-parsed probable cause statements (n=23,742). Triangulated finding: organizational accountability appears in 0.42–0.58% of investigative outputs across both methods.

Kim, S. (2026). *FDA vs NTSB: Cross-Domain Validation of the Convergence Lifecycle.* In preparation.

Five design case studies in a four-layer framework for Regulatory HCI, each grounded in a deployed tool with within-subject user studies (n=20 per prototype), are documented at sorakim-lab.github.io/tools.

Deployed Tools

Twelve production tools deployed at a regulated pharmaceutical site, serving as the empirical substrate for the design research above. All systems operate as unvalidated productivity layers external to validated regulated systems, with team lead awareness and support. Each ships with reproduction guide and team training documentation.

Design Prototypes (5)

- **Procedure Retrieval** — Semantic search over regulated procedural archives. *Layer 1, accessibility.*
- **Cross-Reference Burden** — Traceability viewer for inter-document references in regulated quality documentation. *Layer 2, structure.*
- **Navigation Friction & Loopback** — Behavioral tracker capturing navigation paths, hesitation, and cross-reference jumps. *Layer 2, structure.*
- **Review Friction Visibility** — Prioritized document review dashboard surfacing procedural friction before formal escalation. *Layer 3, coordination.*
- **Root-Cause Reasoning Scaffolder** — Investigative scaffold that preserves analyst judgment while structuring causal reasoning. *Layer 4, reasoning.*

Operational Tools (7)

- **Quality Operations Dashboard** — Unified daily view across stability testing, batch release, and out-of-specification investigations.
- **Change Control Tracker** — Workflow support for electronic change control with action-level burden visibility and extension flags.
- **Reagent Inventory & Expiry Tool** — Automated stock and expiry tracking with rule-based email notifications via Google Sheets and Apps Script.
- **Document Auto-Review Assistant** — AI-based format and compliance pre-check for procedural documents; surfaces drift, does not auto-correct.
- **Project Scheduling Tool** — Morning briefing tool for project managers with backward-derived checkpoints and pre-written check-in messages.
- **Task Manager** — Capacity heatmap embedded in a calendar view, with certification matrix enforced in the assignment interface.
- **Bilingual Document Translation** — Korean–English document production with a team-managed 412-term glossary.

Educational Artifacts

Open access at sorakim-lab.github.io/learning.

- **Using Claude Right: A GMP Quality Lab AI Literacy Series** (2026) — A 15-page AI literacy cookbook for pharmaceutical quality teams. Four chapters covering AI capabilities and limitations; security and data integrity (no credentials, no internal regulated systems, no shared drives; data integrity principles); document writing prompts (procedures, deviations, corrective action, translation); and workflow automation for managers, plus a quick-reference prompt card. Bilingual (English + Korean); PDF.
- **DAY ONE: GMP** (2026) — A pixel-art onboarding game for newcomers entering pharmaceutical regulated environments. Five workday scenarios (gowning, material

verification, procedure retrieval under pressure, deviation reporting, end-of-day sign-off) teach core principles of regulated practice through structured decision points and explanation-after-choice feedback. Built in HTML + JavaScript (Canvas API) with retro pixel typography. Bilingual (English + Korean); web.

Research Corpora & Data Infrastructure

- **FDA Warning Letters Corpus** — 50 letters, 160 violations (2016–2025). The agency's formal regulatory enforcement documents issued when significant compliance failures are identified; used across organizational theory, STS, and computational papers.
- **NTSB Aviation Accident Corpus** — Full Aviation Accident Database (~546 MB; 71,114 findings, 27,877 narratives, 1982–2025) directly analyzed; replication code at `sorakim-lab/ntsb-attribution-analysis`. Methodological contrast with FDA Warning Letters: pharmaceutical regulation makes only final outputs public, leaving the investigative process unobservable; aviation regulation makes the entire investigative pipeline traceable. The cross-domain comparison provides the strongest external validation site for the lifecycle theory developed on FDA data.
- **Cognitive Friction Corpus** — 400 naturally occurring sentences from regulated procedural documents, manually annotated at sentence level.
- **Field Observation Archive** — Field notes from inside a regulated pharmaceutical site (September 2022 – present).

Side Projects

Personal projects outside the main research program. Further detail at sorakim-lab.github.io/projects.

- **Emotion Namer** — A tool that helps name vague feelings precisely. Built on Lieberman's "name it to tame it" research and the Geneva Emotion Wheel (Scherer, 2005); input lands on Russell's valence × arousal plane. Available in Korean, English, and Japanese. *Stack:* React + React Native; Tailwind; i18n; LLM API.
- **Medication Traceability for Crohn's** — A daily tracker for people with chronic illness, linking medication, symptoms, and side effects over time. Health data kept local on-device, exportable as a clean PDF for clinic visits. *Stack:* React Native; Next.js; SQLite + IndexedDB; Recharts; PDF export.
- **Medical Record Reader** — Plain-language translator for clinical reports. Reads test results, produces a plain-language version alongside the original, and flags which findings might be worth asking the doctor about. Explains; does not diagnose. *Stack:* Next.js + React Native; pdf.js + Tesseract OCR; LLM API.
- **Clinical Trial Matcher** — Matches a patient profile against ClinicalTrials.gov inclusion/exclusion criteria, producing a match percentage and surfacing reasons for ineligibility. *Stack:* Next.js + React Native; ClinicalTrials.gov API; spaCy.

- **Terms-of-Service NLP** — Extracts unfair clauses from terms-of-service documents (forced arbitration, unilateral change, broad data-sharing) and tags each with a severity level. *Stack:* Python; clause classification; legal NLP.
- **Urban Equity Collection** — Over four years, ∼21+ data visualizations on urban inequality across six continents (greening, mobility, housing, climate exposure) as a side hobby. *Stack:* geopandas; folium; kepler.gl; OSMnx; ResNet; StreetView CV.
- **AI Usage Taxonomy** — Ongoing qualitative study with quality team colleagues mapping where they use AI, where they trust it, where they don't, and what they specifically watch out for in a regulated environment. *Methods:* Qualitative; semi-structured interviews; grounded coding.

Methods & Technical Skills

- **Research Methods** — Ethnographic fieldwork; document and corpus analysis; within-subject experimental design; think-aloud protocols; behavioral logging; qualitative coding; annotation protocol design; causal reasoning analysis; statistical analysis.
- **Programming & Development** — Python, R; Streamlit (deployment); JavaScript, HTML/CSS; React, Next.js, React Native; Git/GitHub; LaTeX.
- **Text Analysis & NLP** — TF-IDF; sentence-level annotation; document classification; corpus construction; entity linking; temporal embedding; graph extraction; spaCy dependency parsing; rule-based actor extraction.
- **Machine Learning** — PyTorch (CNN, deep learning for image classification); scikit-learn; classification; ensemble methods; unsupervised learning.
- **Regulated-Environment Systems** — Industry-standard pharmaceutical quality management systems; regulated documentation workflows; electronic change control; deviation and corrective action workflows.

Languages

Korean (native) · English (fluent) · Japanese (fluent)